Project Report: Student Attendance System (SAS)

1. Title

Student Attendance System (SAS)

2. Team Members

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3. Problem Statement

Traditional methods of tracking student attendance using paper registers are time-consuming, prone to human error, and make it difficult to perform continuous monitoring. Manually calculating attendance percentages to identify students who fall below the required threshold (defaulters) across multiple subjects is incredibly tedious for administrators. There is a strong need for an automated, centralized, and digitized solution to streamline attendance marking and generate real-time analytical reports.

4. Description

The Student Attendance System (SAS) is a modern, full-stack web application developed to digitalize the attendance management workflow. It empowers administrators and faculty to seamlessly mark, track, and manage student attendance subject-wise and date-wise.

The system features an intuitive Admin Dashboard providing a high-level overview of working days and average attendance. A core feature of the application is the automated **Defaulter Management** engine, which automatically flags any student falling below the mandatory 75% attendance criteria. It supports date-range filtering, and the easy extraction of defaulter lists via PDF and CSV data exports, greatly reducing administrative overhead.

5. Technology Used

Client-Side Scripting (Frontend)

- **Framework/Library:** React.js (via Vite)
- **Styling:** CSS / Styling Libraries
- **Data Visualization & Icons:** Chart.js, React-Icons
- **HTTP Client:** Axios

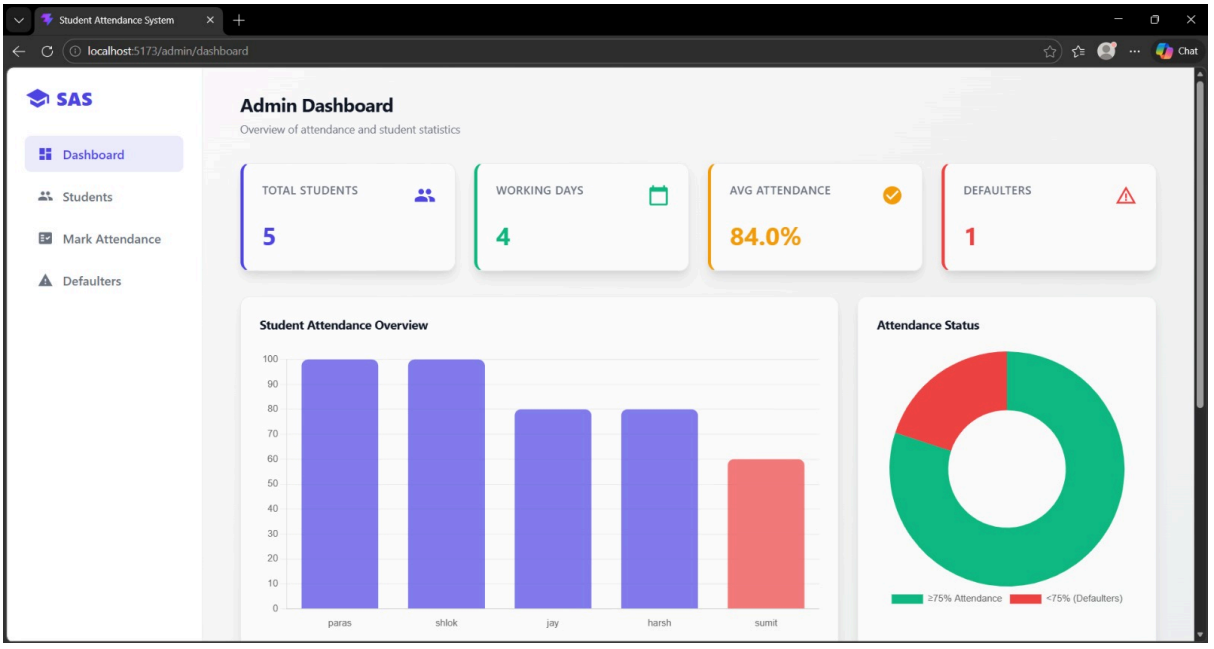
Server-Side Scripting (Backend)

- **Runtime Environment:** Node.js
- **Web Framework:** Express.js
- **Database:** MySQL
- **ORM:** Sequelize

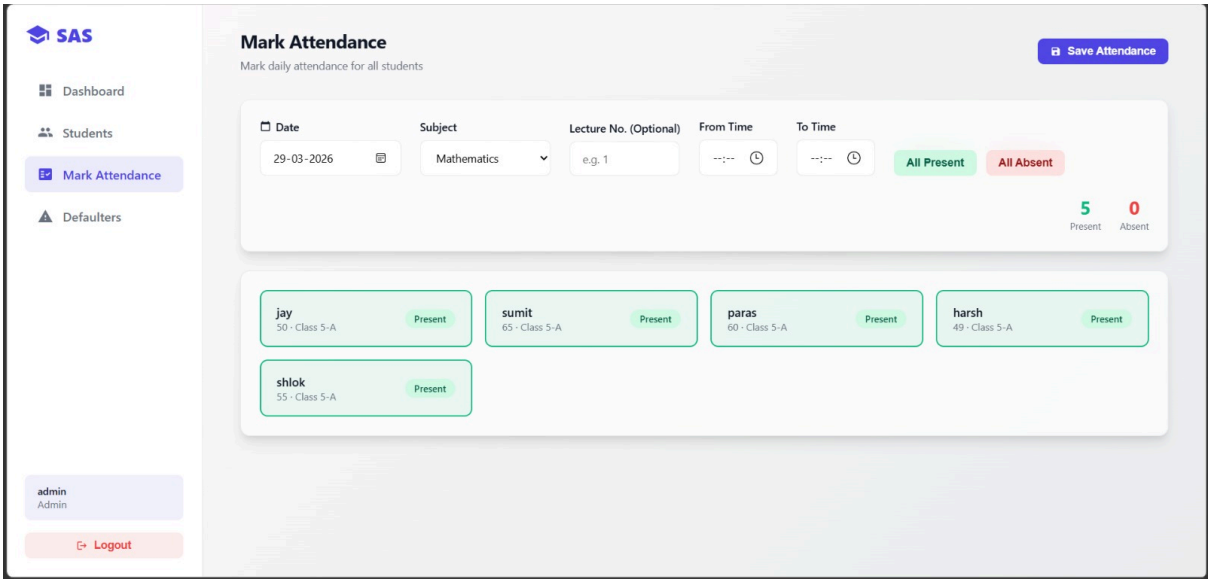
- **Authentication:** JSON Web Tokens (JWT) for secure, role-based access control.

6. Layout Design Snapshots

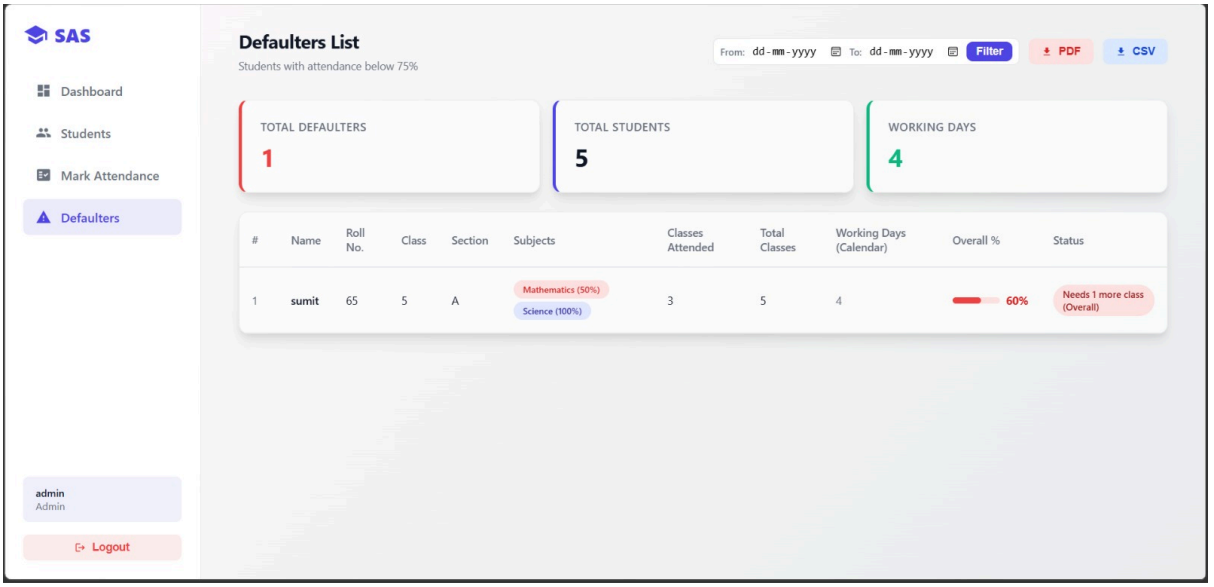
- **Figure 1: Admin Dashboard / Home Page**



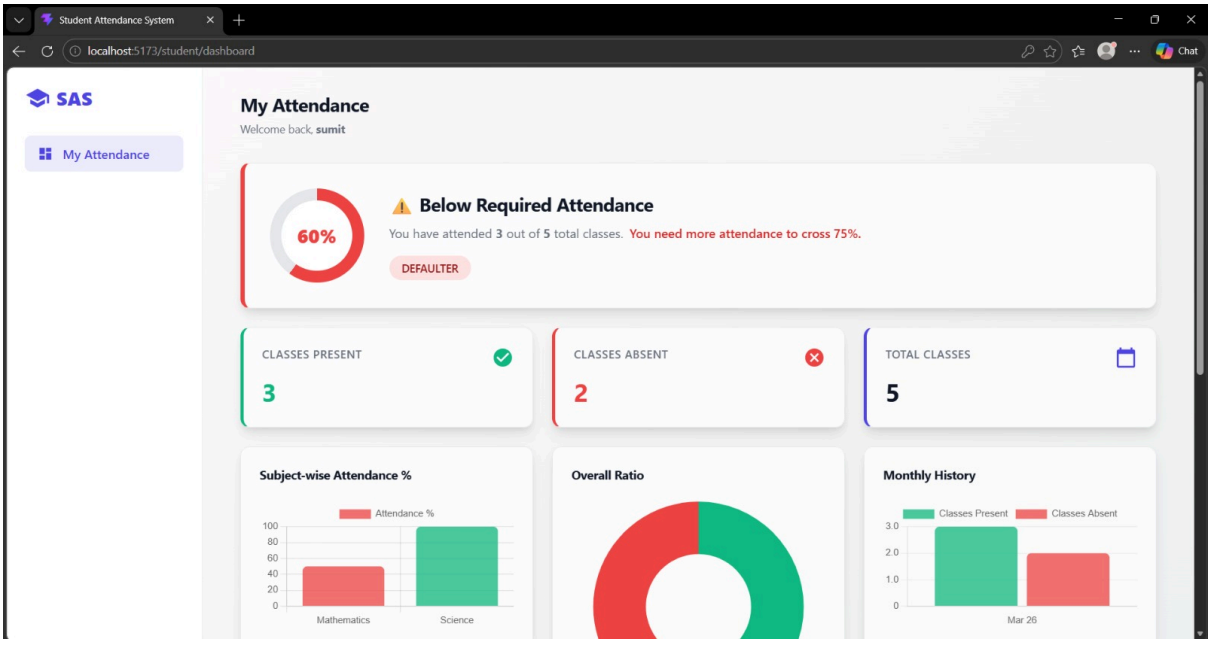
- **Figure 2: Attendance Marking Interface**



- **Figure 3: Defaulter List & Filtering**



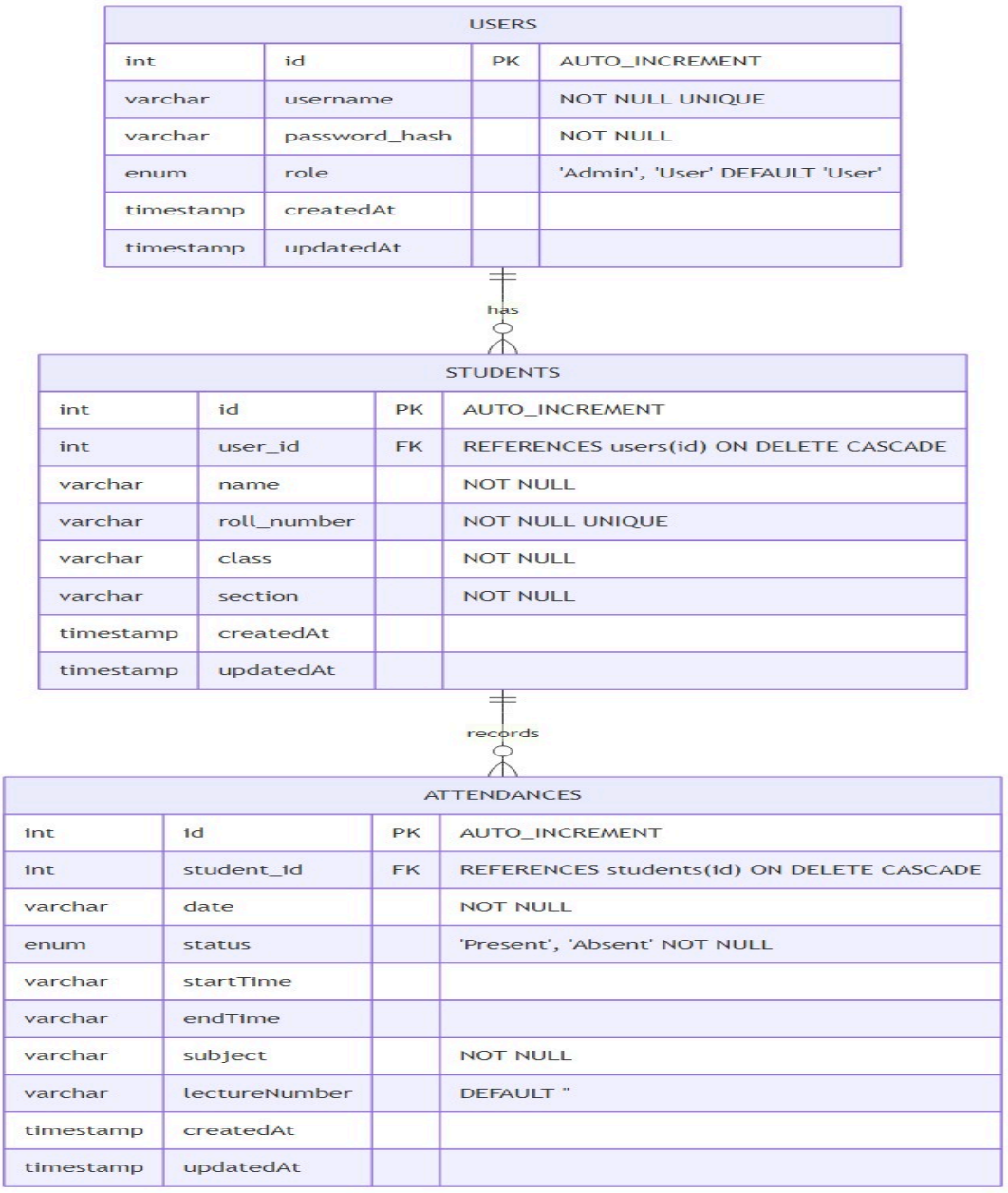
- **Figure 4: Student Dashboard**



7. Data Base Design

The system utilizes a relational database (MySQL) structured to maintain data integrity and efficient querying. The core entities include:

- **Users Table:** Stores administrator credentials and login details (auth).
- **Students Table:** Maintains student profiles (Roll Number, Name, Department, etc.).
- **Attendances Table:** Records individual attendance entries mapped to specific students, subjects, dates, and their present/absent status.



(Demonstration of the Entity-Relationship mapping)

8. Conclusion

The Student Attendance System successfully tackles the inefficiencies associated with manual tracking by providing a robust, automated digital solution. The integration of a responsive React frontend and a reliable Node.js/MySQL backend ensures high performance and data security. By automatically calculating and exporting defaulter lists, the platform significantly reduces processing time for academic staff, allowing them to focus more on educational quality rather than administrative paperwork. Ultimately, the system improves transparency and record accessibility across the institution.